

Applied NLP

DATA 641

Lecture 3 — NLP Pipeline; Evaluation

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Evaluation

Intrinsic Evaluation

Intrinsic evaluation focuses on intermediary objectives. For example, consider a spam-classification system. The ML metric will be precision and recall. Intrinsic evaluation will focus on measuring the system performance using precision and recall.

Metric	Description	Applications
Accuracy	Used when the output variable is categorical or discrete. It denotes the fraction of times the model makes correct predictions as compared to the total predictions it makes.	Mainly used in classification tasks, such as sentiment classification (multiclass), natural language inference (binary), paraphrase detection (binary), etc.
Precision	Shows how precise or exact the model's predictions are, i.e., given all the positive (the class we care about) cases, how many can the model classify correctly?	Used in various classification tasks, especially in cases where mistakes in a positive class are more costly than mistakes in a negative class, e.g., disease predictions in healthcare.
Recall	Recall is complementary to precision. It captures how well the model can recall positive class, i.e., given all the positive predictions it makes, how many of them are indeed positive?	Used in classification tasks, especially where retrieving positive results is more important, e.g., e-commerce search and other information-retrieval tasks.
F1 score	Combines precision and recall to give a single metric, which also captures the trade-off between precision and recall, i.e., completeness and exactness. F1 is defined as $(2 \times \text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$.	Used simultaneously with accuracy in most of the classification tasks. It is also used in sequence-labeling tasks, such as entity extraction, retrieval-based questions answering, etc.
AUC	Captures the count of positive predictions that are correct versus the count of positive predictions that are incorrect as we vary the threshold for prediction.	Used to measure the quality of a model independent of the prediction threshold. It is used to find the optimal prediction threshold for a classification task.
MRR (mean	Used to evaluate the responses retrieved given their	Used heavily in all information-retrieval

Intrinsic Evaluation

Metric	Description	Applications
MAP (mean average precision)	Used in ranked retrieval results, like MRR. It calculates the mean precision across each retrieved result.	Used in information-retrieval tasks.
RMSE (root mean squared error)	Captures a model's performance in a real-value prediction task. Calculates the square root of the mean of the squared errors for each data point.	Used in conjunction with MAPE in the case of regression problems, from temperature prediction to stock market price prediction.
MAPE (mean absolute percentage error)	Used when the output variable is a continuous variable. It is the average of absolute percentage error for each data point.	Used to test the performance of a regression model. It is often used in conjunction with RMSE.
BLEU (bilingual evaluation understudy)	Captures the amount of n-gram overlap between the output sentence and the reference ground truth sentence. It has many variants.	Mainly used in machine-translation tasks. Recently adapted to other text-generation tasks, such as paraphrase generation and text summarization.
METEOR	A precision-based metric to measure the quality of text generated. It fixes some of the drawbacks of BLEU, such as exact word matching while calculating precision. METEOR allows synonyms and stemmed words to be matched with the reference word.	Mainly used in machine translation.
ROUGE	Another metric to compare quality of generated text with respect to a reference text. As opposed to BLEU, it measures recall.	Since it measures recall, it's mainly used for summarization tasks where it's important to evaluate how many words a model can recall.
Perplexity	A probabilistic measure that captures how confused an NLP model is. It's derived from the cross-entropy in a next word prediction task.	Used to evaluate language models. It can also be used in language-generation tasks, such as dialog generation.

Extrinsic Evaluation

- Extrinsic evaluation focuses on evaluating performance on the final objective. On the previous example extrinsic evaluation will focus on measuring the time a user wasted because a spam email went to their inbox or a genuine email went to their spam folder.
- We might ask: if extrinsic evaluation is what matters, why do intrinsic evaluation at all? The reason we must do intrinsic evaluation before extrinsic evaluation is that extrinsic evaluation often includes project stakeholders outside the AI team.
- Intrinsic evaluation can be done mostly by the AI team itself.
- Extrinsic evaluation is a much more expensive process as compared to intrinsic evaluation. Therefore, intrinsic evaluation is used as a proxy for extrinsic evaluation. Only when we get consistently good results in intrinsic evaluation should we go for extrinsic evaluation.
- Also it is worth noting that bad results in intrinsic evaluation often imply bad results in extrinsic evaluation.